

Balanced Health Attainment Score– App Integration Specification

Purpose: Technical requirements for implementing the Balanced Health Attainment Score (BHAS v2.3) within the Balanced Health Institute application. This version updates the biometric metric to Waist-to-Height Ratio (WtHR) and uses binary (all-or-none) scoring for Vitamin D and Vitamin B12.

1. Participant Type Logic

If participant has Type 1 Diabetes:

- Exclude HOMA-IR scoring
- Include Insulin Units/kg/day scoring

Else:

- Include HOMA-IR scoring
- Exclude Insulin Units/kg/day scoring

2. Required Data Inputs (Units)

- Participant Identifier (string)
- Sex (Male/Female) – required for grip scoring
- Type 1 Diabetes (Yes/No) – required for substitution logic
- Fasting Glucose (mg/dL)
- Fasting Insulin (μ U/mL)
- hs-CRP (mg/L)
- Triglycerides (mg/dL)
- HDL (mg/dL)
- Vitamin D 25-OH (ng/mL)
- Vitamin B12 (pg/mL)
- VO2 Max Percentile (age/sex adjusted percentile; numeric 0–100)
- Grip Strength (kg)
- Body Weight (kg)

- Height (cm)
- Waist Circumference (cm)
- Total Daily Insulin Units (units/day) – Type 1 only
- Acute Care Visits (count, past 12 months)
- Advanced Care Plan Status (Yes/No)

3. Derived Calculations

- HOMA-IR = (Fasting Insulin × Fasting Glucose) / 405
- TG/HDL Ratio = Triglycerides / HDL
- Grip Ratio = Grip Strength (kg) / Body Weight (kg)
- Waist-to-Height Ratio (WtHR) = Waist (cm) / Height (cm)
- Insulin Units/kg = Total Daily Insulin Units / Body Weight (kg)

4. Metric Scoring Rules (Points = 1 / 0.5 / 0 unless noted)

- HOMA-IR (Non-Type 1 only): $<1.5 = 1$ | $1.5-2.5 = 0.5$ | $\geq 2.5 = 0$
- hs-CRP: $<1.0 = 1$ | $1.0-3.0 = 0.5$ | $>3.0 = 0$
- TG/HDL: $<2.0 = 1$ | $2.0-3.0 = 0.5$ | $>3.0 = 0$
- Insulin Units/kg (Type 1 only): $<0.6 = 1$ | $0.6-0.8 = 0.5$ | $>0.8 = 0$
- Vitamin D (binary): $>50 \text{ ng/mL} = 1$ | $\leq 50 = 0$
- Vitamin B12 (binary): $>750 \text{ pg/mL} = 1$ | $\leq 750 = 0$
- VO2 Max Percentile: $\geq 60 = 1$ | $40-59 = 0.5$ | $<40 = 0$
- Grip Ratio (Sex-specific): Men $\geq 0.60 = 1$ | $0.50-0.59 = 0.5$ | $<0.50 = 0$; Women $\geq 0.45 = 1$ | $0.35-0.44 = 0.5$ | $<0.35 = 0$
- Waist-to-Height Ratio (WtHR): $\leq 0.50 = 1$ | $0.51-0.56 = 0.5$ | $>0.56 = 0$
- Advanced Care Planning (binary): Documented ACP = 1 | Not documented = 0

5. Total Score Calculation

BHAS Score = Sum of all included metric points.

Each participant has 9 scored categories total:

- Non-Type 1: HOMA-IR + (hs-CRP, TG/HDL, Vit D, B12, VO2, Grip Ratio, WtHR, ACP)
- Type 1: Insulin Units/kg replaces HOMA-IR + (hs-CRP, TG/HDL, Vit D, B12, VO2, Grip Ratio, WtHR, ACP)

Maximum score = 9.0

6. Score Interpretation (Suggested)

8.0–9.0 = Optimal

6.0–7.5 = Healthy

4.0–5.5 = Needs Improvement

0–3.5 = High Risk

7. Tie-Breaker Ranking Logic

If BHAS scores are tied, rank in this order:

- 1) Higher VO2 Max Percentile
- 2) Lower Waist-to-Height Ratio (WtHR)
- 3) Lower hs-CRP
- 4) Fewer Acute Care Visits (past 12 months)

8. App Output Requirements

- Capture and store raw input values with units and timestamps
- Calculate and store derived values (HOMA-IR, TG/HDL, Grip Ratio, WtHR, Insulin Units/kg)
- Assign and store per-metric points (0 / 0.5 / 1; Vitamin D/B12/ACP are binary 0/1)
- Display a user-facing dashboard: raw values + points + BHAS total + category
- Provide population analytics: averages, distributions, % meeting optimal for each metric
- Provide leaderboards with tie-breaker logic
- Export CSV for employer reporting (de-identified where required)